8. Proof

What students need to learn:

Proof by mathematical induction.

To include induction proofs for

(i) summation of series

e.g. show $\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$ or

$\sum_{r=1}^{n} r(r+1) = \dfrac{n(n+1)(n+2)}{3}$

(ii) divisibility

e.g. show $3^{2n} + 11$ is divisible by 4.

(iii) finding general terms in a sequence

e.g. if $u_{n+1} = 3u_n + 4$ with $u_1 = 1$, prove that $u_n = 3^n - 2$.

(iv) matrix products

e.g. show $\begin{pmatrix} -2 & -1 \\ 9 & 4 \end{pmatrix}^n = \begin{pmatrix} 1-3n & -n \\ 9n & 3n+1 \end{pmatrix}$.